Assignment -1

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Sub - AI

Answer-01

PEAS for smart vaccum cleaner agent-
PEAS describes the Task environment of an intelligent agent.

- Performance- Maximum cleanliness, minimum time, low pressure consumption avoid obstacles.
- Environment - Rooms, furniture, dirt, pets humans
- Actuators - Wheels vaccum motor, brushes display speaker.
- Sensors - Dirt sensor obstacle sensor, camera, infrared sensor, battery

Answer- 02

8- Puzzle Problem

a) State Representation -

The state is represented by a 3X3 grid containing numbered tiles (1-8) and one blank space. This is overall and about of space state representation

(b) Initial State goal state Actions -
BFS is a uniformed search algo
initial state :- Starting arrangements of tiles
Goal State :- Tiles arranged in order with bank at the last position
Actions :- Move the blank tile up down, left, right when it should be possible.

Answer-03

State space representation - It is the set of all possible state of a problem for the initial state to the goal state. It is usually represented as a graph, where nodes are states and Edge are action.

Breadth - First search (BFS) :-
BFS is a uniformed search algorithm that explode no dey level by level and guarantee the shortest path.

Python program for BFS -

```
From collections import deque
graph = {
        'A' = ['B', 'C'],
        'B' = ['D', 'E'],
        'C' = ['F'],
        'D' = [],
        'E' = ['F']
        'F' = []
    }

def bfs (start, goal):
        visited = set()
        queue = deque ([[ start ]])
        while queue:
                path = queue.popleft()
                node = path [-1]
                if node == goal:
                        return path

        if node not in visited:
                visited node (add)
        for neighbour in graph [node]:
```

```
new_path = list(path)
new_path = append(neighbour)
queue.append(new_path)
print("path :", bfs('A', 'F'))
```

Output :-

Path : A → C → F